

— $\varepsilon_{\text{relic}}$ Boltz-eq

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$m_\chi = 0.6 m_{A'}$
 $\Omega_{DM} = 0.265$

$\alpha_D = 0.1$

$\alpha_D = 0.03$

$\alpha_D = 0.95$

ε

10^{-7}
 10^{-2}

10^{-1}

10^0

$m_{A'} \text{ (GeV)}$

